Report
Of

Land Degradation Prediction System

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By
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Name of Student: DEEPAYAN BISWAS
Batch: 2025-29

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CANDIDATE’S DECLARATION

I, DEEPAYAN BISWAS do hereby declare that the internship report titled “Land Degradation Prediction System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Deepayan Biswas

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Deepayan Biswas

Roll No.: 25SCS1003001464

INTERNSHIP COMPLETION CERTIFICATE

MARS

CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/034

Date of Issue: July 31, 2026

This is to certify that

Deepayan Biswas

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

Manish Prateek

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Land is one of the most fundamental natural resources, supporting agriculture, biodiversity, and human settlement. Over the past few decades, rapid urbanization, deforestation, unsustainable agricultural practices, and climate variability have accelerated the degradation of land quality across many regions of India, including the state of Uttar Pradesh. Left unchecked, land degradation reduces agricultural productivity, disrupts local ecosystems, and increases vulnerability to droughts and soil erosion.

Traditional methods of assessing land degradation rely heavily on manual field surveys and visual interpretation of satellite imagery, which are time-consuming, subjective, and difficult to scale across large geographic areas. With the increasing availability of remote-sensing-derived environmental data, machine learning offers a faster and more consistent alternative for identifying and classifying degraded land at scale.

Land Use Land Cover (LULC) classifies the earth's surface into categories — forest, agriculture, built-up, water, and barren land — using satellite-derived indicators, and forms the basis on which land degradation is assessed in this project. My approach, instead of relying on manual visual inspection, uses a trained machine learning model to automatically classify degradation severity from environmental data.

4.2 Organization Profile

This internship was undertaken with MARS – Centre for Sustainable Community Development, an organisation focused on sustainable community development initiatives. The internship was carried out under the MARS Internship Program (GREEN-AG Track), which centres on projects at the intersection of environmental sustainability, land management, and agriculture. As part of this track, the project applied data science and machine learning methods to a real-world environmental problem — classifying land degradation severity — in support of the organisation's broader mission of promoting sustainable land use and community-driven environmental stewardship.

4.3 Problem Statement

Given land cover fractions and climate/soil indicators for a grid cell in Uttar Pradesh, the goal is to predict the severity of land degradation — Low, Moderate, or High — so that restoration efforts can be prioritized efficiently and objectively, rather than relying on ad-hoc manual assessment.

4.4 Project Objectives

The primary objective of this project is to build and deploy a machine-learning system that can automatically classify land degradation severity across Uttar Pradesh. Specific objectives include:

1. To aggregate land cover, climate, and soil data into a uniform 5 km × 5 km grid across 20 districts of Uttar Pradesh.
2. To design a Land Degradation Index (LDI) that quantifies degradation severity from environmental indicators.
3. To train and tune a classification model capable of predicting degradation severity (Low, Moderate, High) with high confidence.
4. To identify which environmental features most strongly influence degradation, using explainability techniques (SHAP).
5. To deploy the model as an accessible platform supporting single-cell prediction, batch prediction, EDA, and geographic visualization for real-world use by authorities and communities.

4.5 Scope of the Project

Attribute	Detail
State	Uttar Pradesh, India
Districts covered	20 (including Chandauli, Badaun, Banda, Aligarh, and others)
Grid resolution	5 km × 5 km
Time period	2020–2024
Total observations	22,645 grid cells

A wide mix of cropland, forest, grassland, and built-up land is represented across the 20 districts, along with diverse hydrology and soil-moisture conditions across the Gangetic plain. This state-wide grid gives the model enough variety in land cover and climate to generalize degradation patterns across districts. The scope of the project is limited to Uttar Pradesh and the 2020–2024 period; extension to other states or time periods is identified as future work.

4.6 Technologies and Tools Used

Layer	Technology
Programming Language	Python
Machine Learning	scikit-learn (Logistic Regression, preprocessing, model tuning)
Explainability	SHAP (SHapley Additive exPlanations)
Backend / Deployment	Flask
Data Handling	Pandas, NumPy
Visualization	Matplotlib / Plotly (EDA and geographic dashboards)

4.7 System Architecture

The system follows a pipeline architecture that moves from raw environmental data to a deployed prediction interface:

1. Data Layer — Land cover fractions (Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland) and climate/soil variables (NDVI, Rainfall, Temperature, Soil Moisture) are sourced per grid cell.
2. Aggregation Layer — Raw indicators are aggregated into a uniform 5 km × 5 km grid across the 20 districts, producing 22,645 grid-cell observations.
3. Feature Engineering Layer — 13 environmental features are combined with one-hot encoded district identity (19 features), giving 32 total input features per grid cell.
4. Modelling Layer — A Logistic Regression classifier is trained and tuned on these features to predict the Low / Moderate / High degradation class, with SHAP used to explain feature contributions.
5. Application Layer — The trained model and preprocessing artefacts are served through a Flask backend, exposing six modules: Single Prediction, Batch Prediction, Model Performance, EDA Dashboard, SHAP Explainability, and Geographic Dashboard.

4.8 Methodology

Data Sources:

Category	Details
Land Cover Fractions	Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland (8 classes per grid cell)

Climate & Soil Variables	NDVI, Rainfall, Temperature, Soil Moisture (averaged per grid cell)

Workflow:

1. Aggregate land cover fractions and climate/soil variables into 5 km × 5 km grid cells.
2. Compute a Land Degradation Index (LDI) per cell from these environmental indicators.
3. Label each cell's LDI into Low / Moderate / High degradation classes.
4. Train and tune a Logistic Regression classifier (13 environmental + 19 district features).
5. Validate performance and interpret feature influence using SHAP explainability.

Degradation Severity Classification — every grid cell is assigned one of three degradation classes:

Class	Description
Low	Healthy vegetation cover, stable soil moisture, minimal bare land or built-up encroachment.
Moderate	Noticeable vegetation loss or rising bare/built-up share; land under stress but recoverable.
High	Severe vegetation loss, high bare land, low soil moisture — urgent restoration needed.

Model Selected — a Tuned Logistic Regression model was selected for deployment, trained on 13 environmental features plus district identity. It outputs the Low / Moderate / High class with a confidence score for every 5 km × 5 km grid cell.

Attribute	Detail
Model	Tuned Logistic Regression
Task	Multiclass classification
Classes	Low, Moderate, High
Features	13 environmental + 19 district (one-hot)

4.9 Expected Outcomes

- A Tuned Logistic Regression model that classifies land degradation (Low/Moderate/High) across 22,645 grid cells in 20 Uttar Pradesh districts.
- A deployed platform offering single-cell and batch prediction, model performance evaluation, EDA, SHAP explainability, and geographic visualization — tested end-to-end on the full 22,645-row dataset.
- A clear view of which environmental features (e.g. temperature, rainfall, soil moisture) most strongly influence predicted degradation severity, supporting more targeted restoration planning.
- A reusable foundation that can be extended to additional states, automated satellite data ingestion, and mobile access for field officers in future iterations.

4.10 Certificates of Completion and Communication Mail Proof

The Internship Completion Certificate issued by MARS – Centre for Sustainable Community Development is provided in the preceding section of this report (Internship Completion Certificate). Any additional official communication correspondence related to this internship (offer confirmation, extension, or completion emails) should be appended here as supporting proof where available.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

The following AI tools and platforms were used during the development of the Land Degradation Prediction System, primarily for building and refining the web platform:

- **Claude AI —** used for code assistance and development support.
- **Google Antigravity —** used for platform/UI development assistance.
- **ChatGPT —** used for code assistance and development support.
- **Stitch AI —** used for UI/design assistance.

Additional technical references:

1. Scikit-learn Documentation — Logistic Regression, Model Tuning and Evaluation. https://scikit-learn.org/

2. SHAP (SHapley Additive exPlanations) Documentation — Model Explainability. https://shap.readthedocs.io/

3. Flask Documentation — Web Application Backend. https://flask.palletsprojects.com/